BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Defensive Design and Maintainability

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does authentication establish?

2. What is POP used for?

A) Sending new email
B) Transferring web pages
C) Looking up IP addresses
D) Downloading email

3. "Changing one protocol layer always requires every other layer to be redesigned." Correct this. Explain why.

4. A drive with private records is lost. "Encryption stops files being stolen or deleted." Correct this. What does it protect?

Today's objectives

Explain defensive design and maintainable code. Apply this to another software scenario.

Key vocabulary - Read, cover, recall

Meaningful naming

Descriptive identifiers such as totalCost make code easier to understand.

Indentation

Consistent indentation reveals which instructions belong in a block.

Comments

Explain important intent, rather than repeating obvious syntax line by line.

Subprograms

Splitting code into related functions or procedures helps reuse, testing and maintenance.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Meaningful naming	Descriptive identifiers such as totalCost make code easier to understand. Consider this in a library software.
Indentation	Consistent indentation reveals which instructions belong in a block. Consider this in a ticketing program.
Comments	Explain important intent, rather than repeating obvious syntax line by line. Consider this in a school marking tool.
Subprograms	Splitting code into related functions or procedures helps reuse, testing and maintenance. Consider this in a shop checkout.

I Do - One worked example

A library software needs meaningful naming and indentation. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1].

3. Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1].

4. Give a specific point and a linked reason for each feature.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a ticketing program, explain the purpose of meaningful naming. [2 marks]

2. A school marking tool needs indentation. Explain its role. [2 marks]

3. A shop checkout uses comments and subprograms. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Add a comment explaining what the subtraction calculates.

```
originalKB = 500
compressedKB = 300
// add your comment here
savedKB = originalKB - compressedKB
print(savedKB)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how indentation and comments help in a travel app. [4 marks]

Q4. Explain the roles of meaningful naming, comments and subprograms in a club scoring. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Descriptive identifiers such as totalCost make code easier to understand. ☐ T ☐ F
2. Descriptive identifiers such as totalCost make code easier to understand ☐ T ☐ F
3. Consistent indentation reveals which instructions belong in a block ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Whether a user has supplied suitable evidence of the claimed identity. ☐

Q2 D - Downloading email ☐

Q3 Layers split communication into tasks. They provide services to each other and help compatibility, development and finding faults. ☐

Q4 Without the key, an attacker cannot easily read ciphertext. Encryption does not stop theft or deletion. Use other controls for other threats. ☐

Code First - Model

```
originalKB = 500
compressedKB = 300
// Work out how much storage compression saved.
savedKB = originalKB - compressedKB
print(savedKB)
```

Output: 200

We Do - Check the model

1. Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1]. ☐

2. Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1]. ☐

3. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. Subprograms [1]; Splitting code into related functions or procedures helps reuse, testing and maintenance. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Meaningful naming [1]; Comments [1]. ☐

Q3 Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1]. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. ☐

Q4 Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1]. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. Subprograms [1]; Splitting code into related functions or procedures helps reuse, testing and maintenance. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: